$$^{n}C_{r} = \frac{\lfloor\underline{n}}{\lfloor\underline{r}\,\lfloor\underline{n-r}}, \quad ^{n}C_{r} = {}^{n}C_{n-r}.$$

$$^{n}C_{1} = \frac{\lfloor\underline{n}}{\lfloor\underline{1}\,\lfloor\underline{n-1}} = \frac{n\lfloor\underline{n-1}}{1\cdot\lfloor\underline{n-1}} = n$$

$$^{n}C_{2} = \frac{\lfloor\underline{n}}{\lfloor\underline{2}\,\lfloor\underline{n-2}} = \frac{n(n-1)\lfloor\underline{n-2}}{2\cdot 1\,\lfloor\underline{n-2}} = \frac{n(n-1)}{2}.$$

$$^{n}C_{n} = \frac{\lfloor\underline{n}}{\lfloor\underline{n}\,\lfloor\underline{n-n}} = \frac{\lfloor\underline{n}}{\lfloor\underline{n}\cdot\lfloor\underline{0}} = \frac{1}{1} = 1.$$

Eg. $^{5}C_{2} = {}^{5}C_{5-2} = {}^{5}C_{3}.$

H.W

* $y = (\sin^{-1}x)^2$, then show that (b)

i) $(1-x^2)y_2 - xy_1 = 2$

ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

* If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$. Prove that

i) $x^2y_2 + xy_1 + n^2y = 0$

ii) $x^2y_{n+2} + (2n+1)xy_{n+1} + 2n^2y_n = 0$

Here $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$

$= n(\log x - \log n)$

$\therefore -\frac{1}{\sqrt{1-\frac{y^2}{b^2}}} \cdot \frac{d}{dx}\left(\frac{y}{b}\right) = n \cdot \frac{1}{x}$

$\Rightarrow -\frac{b}{\sqrt{b^2-y^2}} \cdot \frac{1}{b} \cdot y_1 = \frac{n}{x}$

$\Rightarrow x^2y_1^2 = n^2(b^2-y^2)$.

$\therefore x^2 \cdot 2y_1y_2 + 2xy_1^2 = n^2(-2yy_1)$

$\Rightarrow x^2y_2 + xy_1 + n^2y = 0$. Proved

Differentiate it n times by Leibnitz's theorem

$$\left[(1+x^2)y_{n+2} + {}^nC_1 \cdot 2x\, y_{n+1} + {}^nC_2 \cdot 2 \cdot y_n\right]$$
$$+ \left[x y_{n+1} + {}^nC_1 \cdot 1 \cdot y_n\right] - m^2 y_n = 0$$

$$= (1+x^2)y_{n+2} + 2nx y_{n+1} + \frac{n(n-1)}{2} \cdot 2y_n$$
$$+ x y_{n+1} + n y_n - m^2 y_n = 0$$

$$= (1+x^2)y_{n+2} + 2nx y_{n+1} + n(n-1)y_n$$
$$+ x y_{n+1} + n y_n - m^2 y_n = 0$$

$$= (1+x^2)y_{n+2} + (2n+1)x y_{n+1} + (n^2-n+n-m^2)y_n = 0$$

$$= (1+x^2)y_{n+2} + (2n+1)(x y_{n+1} + (n^2-m^2)y_n = 0.$$

Proved

* leibnitz's theorem

U and V are two function of x and all their derivatives exist, then by leibnitz's theorem

$$D^n(UV) = (D^nU)V + {}^nC_1(D^{n-1}U)(DV) + {}^nC_2(D^{n-2}U)(D^2V) + \cdots + {}^nC_r(D^{n-r}U)(D^rV) + \cdots + U(D^nV).$$

$$\text{or } (UV)_n = U_nV + {}^nC_1U_{n-1}V_1 + {}^nC_2U_{n-2}V_2 + {}^nC_3U_{n-3}V_3 + \cdots + {}^nC_rU_{n-r}V_r + \cdots + UV_n.$$

Now differentiate this equation n times by leibnitz's theorem.

$$\left[x^2 y_{n+2} + {}^nC_1 \cdot 2x\, y_{n+1} + {}^nC_2\, 2 y_n\right] + \left[x y_{n+1} + {}^nC_1 \cdot 1 \cdot y_n\right] + n^2 y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + 2nx\, y_{n+1} + \frac{n(n-1)}{2} \cdot 2 y_n + x y_{n+1} + n y_n + n^2 y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + 2nx\, y_{n+1} + n(n-1) y_n + x y_{n+1} + n y_n + n^2 y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + (2n+1)\, x y_{n+1} + (n^2 - n + n + n^2) y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + (2n+1)\, x y_{n+1} + 2n^2 y_n = 0$$

Proved

We have shown that

$$(1-x^2)y_2 - xy_1 + m^2 y = 0$$

Differentiate it n times w.r.to x

$$\left[(1-x^2)y_{n+2} + {}^nC_1(-2x)y_{n+1} + {}^nC_2(-2)y_n\right]$$

$$- \left[xy_{n+1} + {}^nC_1 \cdot 1 \cdot y_n\right] + m^2 y_n = 0$$

or, $(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2} \cdot 2y_n$

$$- xy_{n+1} - ny_n + m^2 y_n = 0$$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-n)y_n - ny_n + m^2 y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-n+n)y_n + m^2 y_n = 0$

or $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-m^2)y_n = 0.$

Proved

* H.W

If $\cos(m\sin^{-1}x)$, then Prove that

$$(1-x^2)y_{n+2} + (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0.$$

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H.W

* If $y = \tan^{-1} x$ then show that

i) $(1+x^2) y_2 + 2x y_1 = 0$

ii) $(1+x^2) y_{n+2} + 2(n-1) x y_{n+1} + n(n+1) y_n = 0.$

* If $y = \{x + \sqrt{1+x^2}\}^m$, show that

$(1+x^2) y_{n+2} + (2n+1) x y_{n+1} + (n^2 - m^2) y_n = 0.$

Here $y = \{x + \sqrt{1+x^2}\}^m$

$\therefore y_1 = m\{x + \sqrt{1+x^2}\}^{m-1} . \left\{1 + \dfrac{2x}{2\sqrt{1+x^2}}\right\}.$

$\Rightarrow y_1 = m . \dfrac{(x+\sqrt{1+x^2})^m}{\cancel{(x+\sqrt{1+x^2})}} . \dfrac{\cancel{(x+\sqrt{1+x^2})}}{\sqrt{1+x^2}}$

$= \dfrac{my}{\sqrt{1+x^2}}$

$\Rightarrow \sqrt{1+x^2} . y_1 = my.$

$\Rightarrow (1+x^2) . y_1^2 = m^2 y^2.$

Differentiate again w.r. to x.

$(1+x^2) . 2 y_1 y_2 + 2x . y_1^2 = m^2 . 2 y y_1$

$\Rightarrow (1+x^2) y_2 + x y_1 - m^2 y = 0.$

Proved

* If $y = e^{\tan^{-1}x}$ then show that

$$(1+x^2)y_2 + (2x-1)y_1 = 0$$

and $(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$

Here $y = e^{\tan^{-1}x}$.

$$\therefore y_1 = e^{\tan^{-1}x} \cdot \frac{d}{dx}(\tan^{-1}x)$$

$$= y \cdot \frac{1}{1+x^2}.$$

or $(1+x^2)y_1 = y$

or $(1+x^2)y_2 + (0+2x)y_1 = y_1$

or $(1+x^2)y_2 + 2xy_1 = y_1$

or $(1+x^2)y_2 + (2x-1)y_1 = 0$.

Differentiate it n times w.r.to x

$$[(1+x^2)y_{n+2} + {}^nC_1 \cdot 2x \cdot y_{n+1} + {}^nC_2 \cdot 2 \cdot y_n] + [(2x-1)y_{n+1} + {}^nC_1 \cdot 2 \cdot y_n] = 0$$

or $(1+x^2)y_{n+2} + 2nxy_{n+1} + 2\cdot\frac{n(n-1)}{2}y_n + (2x-1)y_{n+1} + 2ny_n = 0$

or $(1+x^2)y_{n+2} + \{2(n+1)x - 1\}y_{n+1} + (n^2 - n + 2n)y_n = 0$

or $(1+x^2)y_{n+2} + \{2(n+1)x - 1\}y_{n+1} + n(n+1)y_n = 0$

Proved

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* If $y = \sin(m\sin^{-1}x)$ then show that
$(1-x^2)y_2 - xy_1 + m^2y = 0$

then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-m^2) = 0$

Here $y = \sin(m\sin^{-1}x)$

$\therefore y_1 = \cos(m\sin^{-1}x) \cdot \frac{d}{dx}(m\sin^{-1}x)$

$= \cos(m\sin^{-1}x) \frac{m}{\sqrt{1-x^2}}$.

or, $\sqrt{1-x^2} \cdot y_1 = m\cos(m\sin^{-1}x)$

Squaring both sides w.r. to x

$(1-x^2)y_1^2 = m^2\cos^2(m\sin^{-1}x)$

$= m^2\{1-\sin^2(m\sin^{-1}x)\}$

$= m^2(1-y^2)$

Differenting both sides w.r. to x

$(1-x^2) \cdot 2y_1y_2 + y_1^2(-2x) = m^2(0-2yy_1)$

Cancelling ~~both sides~~ the factor $2y_1$ from both sides.

$\underline{(1-x^2)y_2} - \underline{xy_1} + m^2y = 0$ proved

(3)

* If $x = \sin\left(\frac{1}{a}\log y\right)$, then show that $(1-x^2)y_2 - xy_1 - a^2y = 0$, Also show that

$$(1-x^2)\,y_{n+2} - (2n+1)\,xy_{n+1} - (n^2+a^2)\,y_n = 0$$

Here $x = \sin\left(\frac{1}{a}\log y\right)$.

$$\therefore \frac{1}{a}\log y = \sin^{-1}x$$

$$\therefore \log y = a\sin^{-1}x$$

$$\therefore y = e^{a\sin^{-1}x}.$$

Differentiating both sides w.r.to x

$$y_1 = \underline{e^{a\sin^{-1}x}}\cdot\frac{d}{dx}\left(a\sin^{-1}x\right)$$

$$= y\cdot\frac{a}{\sqrt{1-x^2}}.$$

$$\therefore \sqrt{1-x^2}\cdot y_1 = ay \quad \text{[Squaring both sides]}$$

$$\therefore (1-x^2)\,y_1^2 = a^2y^2$$

Again differentiating w.r.to x

$$(1-x^2)\cdot 2y_1y_2 + y_1^2(-2x) = a^2\cdot 2y\cdot y_1$$

Cancelling $2y_1$ from both sides

$$(1-x^2)\,y_2 - xy_1 - a^2y = 0$$

Proved

We have shown that

$$(1-x^2)y_2 - xy_1 - a^2y = 0$$

Differentiate it n times w.r.to x

$$\left[(1-x^2)y_{n+2} + {}^nC_1(-2x)y_{n+1} + {}^nC_2(-2)y_n\right] - \left[xy_{n+1} + {}^nC_1\cdot 1\cdot y_n\right] - a^2y_n = 0$$

or, $$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2}\cdot 2y_n - xy_{n+1} - ny_n - a^2y_n = 0$$

or $$(1-x^2)y_{n+2} - 2nxy_{n+1} - (n^2-n)y_n - xy_{n+1} - ny_n - a^2y_n = 0$$

or $$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-n+n+a^2)y_n =$$

or $$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

* If $y = a\cos(\log x) + b\sin(\log x)$, show that

i) $x^2 y_2 + xy_1 + y = 0$

ii) $x^2 y_{n+2} + (2n+1)\, x y_{n+1} + (n^2+1)\, y_n = 0$.

Here $\cancel{x^2 y_2 + xy_1 + y = 0}$

$\therefore\ y = a\cos(\log x) + b\sin(\log x)$.

$\therefore\ y_1 = \left[-a\sin(\log x) + b\cos(\log x)\right] \cdot \frac{1}{x}$

$\Rightarrow\ xy_1 = -a\sin(\log x) + b\cos(\log x)$.

Differentiate again w.r. to x

$xy_2 + y_1 = -\frac{1}{x}\left[a\cos(\log x) + b\sin(\log x)\right]$

$= -y/x$

$\therefore\ x^2 y_2 + xy_1 + y = 0$ <u>Proved</u>

Now differentiate it n times by Leibnitz's theorem.

$\left(x^2 y_{n+2} + {}^{n}C_1 \cdot 2x \cdot y_{n+1} + {}^{n}C_2 \cdot 2 y_n\right) + \left(x y_{n+1} + {}^{n}C_1 \cdot 1 \cdot y_n\right) + y_n = 0$

$\Rightarrow\ x^2 y_{n+2} + 2nx\, y_{n+1} + \frac{n(n-1)}{2} \cdot 2y_n + xy_{n+1} + ny_n + y_n = 0$

$\Rightarrow\ x^2 y_{n+2} + (2n+1)\, x y_{n+1} + (n^2 - n + n + 1)\, y_n = 0$

$\Rightarrow\ x^2 y_{n+2} + (2n+1)\, x y_{n+1} + (n^2+1)\, y_n = 0$.

<u>Proved</u>